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## 1 Implementation Plan: the SeisSol v3.4.1 RSSRW PREFERRED THERMAL CASE2 problem in SEAS-MFEM, with mixed flux

Date: 2026-07-08 Author: exploration + plan pass over seas-mfem-spatial-dyn-driver, branch system/spatial\_dyn\_driver

Successor to the v3.0.0 port plan (2026-06-19), in project\_7.0\_alternative/document/.

### 1.1 Overview

Run the **same physical problem** as the SeisSol deck

~/Downloads/seisol\_quakeworx/safs\_seisol\_v3\_4\_1\_RSSRW\_PREFERRED\_THERMAL\_CASE2/

through seas\_spatial\_dyn\_driver with MFEM's **mixed (adjacent-central) flux + explicit RK**, which SeisSol does not have. Same mesh, same CVM material, same Andersonian  $k=1.8$  initial stress, same temperature-zoned rate-and-state + strong-rate-weakening friction, same 75 MPa /  $R=2000$  m compact nucleation bump, same 150 s of simulated time.

This is **not** a bit-for-bit reproduction: SeisSol runs ADER-DG order 4 ( $p=3$ ) with clustered LTS and a pure Godunov flux; MFEM runs  $p=1$  global time stepping with a central flux on fault-adjacent faces. The *inputs* transfer exactly; the *discretization* is the experimental variable.

## The experiment (user directive, 2026-07-09): mixed flux on the large-rv mesh

Two deliberate deviations from the pristine v3.4.1 deck, both in service of one question — **can MFEM’s mixed (adjacent-central) flux numerically tolerate a large-rv mesh that SeisSol had to fix geometrically?**

- **Mesh: keep the un-fixed deep19km mesh** (`safv4_deep19km_sub500m_flattop_nwtrim_fixed_opt`), **not** the `v3_5_3_rvfix4` mesh. Large *rv* (Zhang 2023  $rv > 3$ ) drives a tensile flip  $\rightarrow$  zero strength  $\rightarrow$  runaway under pure upwind; SeisSol’s answer was a geometry fix (`rvfix4`). Mixed flux is the Zhang-2023  $\sigma_n$ -leak cure at the *numerical* level, so the point is to run it on the un-fixed mesh and see whether it holds.
- **Stress:  $k=1.5$  + freeze500m** (the SeisSol-verified `v3_5_3` field, a mesh-independent grid), swapped in for the deck’s  $k=1.8$  so shallow free-surface facets can’t free-slip at near-zero normal stress (`freeze500m`). See Phase 5.

So: **deep19km (large-rv) mesh + `v3_5_3`  $k=1.5$ -freeze500m stress + mixed flux**. Do not swap in `rvfix4`.

### Headline

**Four of the six input channels already work.** The mesh is already in the repo in MFEM-readable form and is provably the same mesh SeisSol ran. Stress and material have existing `.nc`  $\rightarrow$  `data_projection_v1` HDF5 converters that need a re-run, not a rewrite. The flux/integrator combination, the FL=103 friction law, and the compact-bell nucleation are all implemented and unit-tested.

**Two things genuinely block the run, and both are small:**

| #         | Blocker                                                                                                                                                                    | Fix size                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>B1</b> | No way to feed a 3-D $rs_a(x, y, z)$ / $rs_{srW}(x, y, z)$ field to the friction resolver. The thermal deck’s entire novelty (CTM temperature zoning) lives in that field. | $\sim 120$ lines C++ +<br>$\sim 150$ lines Python + 3 tests |
| <b>B2</b> | <code>f_w_default</code> parser guard rejects $f_w = 0.0$ , which is exactly what $RS_{\mu W} = 0.0$ in the deck asks for.                                                 | 1 line + 1 test                                             |

Everything else is data preparation, config authoring, and one optional new output writer.

## 1.2 Background – verified findings

Every claim below was checked against the source or the files themselves during this exploration.

### The deck (SeisSol side)

`parameters.par` + `safs_fault.yaml` + `safs_initial_stress.yaml` + `safs_material_cvm.yaml`:

| Quantity          | Value                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Friction law      | FL = 103 (rate-and-state slip law + strong rate weakening)                                                                                                  |
| RS_f0 / RS_sr0    | 0.6 / 1e-6                                                                                                                                                  |
| RS_b              | 0.019 (constant — v1.1.3 does not read spatial rs_b)                                                                                                        |
| RS_muW (f_w)      | 0.0                                                                                                                                                         |
| rs_sl0 (Dc)       | 0.10 m (constant)                                                                                                                                           |
| RS_iniSlipRate1/2 | 1e-12 / 0.0                                                                                                                                                 |
| rs_a, rs_srW      | <b>spatial</b> , from<br>safs_friction_thermal_case2.nc (ASAGI)                                                                                             |
| Nucleation        | Tnuc_s = 75e6 · exp(r <sup>2</sup> /(r <sup>2</sup> −R <sup>2</sup> )), R = 2000<br>m, centre (609062.8722, 3709528.1324,<br>−10000.0), Tnuc_n = Tnuc_d = 0 |
| Nucleation ramp   | s_0 = 0, t_0 = 1.0 s (SeisSol smoothStep)                                                                                                                   |
| Fault reference   | XRef, YRef, ZRef = (0, −1, 0), refPointMethod<br>= 1 (reference <b>vector</b> )                                                                             |
| Initial stress    | safs_stress_andersonian_k1.8.nc,<br>Andersonian C1 k=1.8, hydrostatic Pp <b>already<br/>removed</b> (effective), compression <b>NEGATIVE</b>                |
| Material          | safs_material_cvm.nc, compound {rho, mu,<br>lambda}                                                                                                         |
| Mesh              | mesh_preferred.puml.h5, 1,241,419 tets /<br>250,588 vertices, 172,593 fault facets                                                                          |
| Discretization    | CFL = 0.5, ClusteredLTS = 2, order 4<br>(compile-time:<br>SeisSol_Release_drome_4_elastic)                                                                  |
| Flux              | numflux = numfluxnearfault = 'godunov'<br>(pure upwind everywhere)                                                                                          |
| EndTime           | 150.0 s                                                                                                                                                     |
| Plasticity        | off                                                                                                                                                         |

Sidcar grids (all ASAGI: NETCDF4, one compound var data, dims (z,y,x), float32 members, float64 strictly-increasing equidistant axes, z = elevation):

| file             | dims (z,y,x)       | $\Delta x, \Delta y, \Delta z$ [m] | extent [m]                                       | members                                             |
|------------------|--------------------|------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| friction (CASE2) | 107 × 194 ×<br>263 | 1500, 1500, 200                    | x[303,696]k<br>y[3612,3901.5]k<br>z[−21000,+200] | rs_a ∈ [0.015,<br>0.0332]; rs_srW<br>∈ [0.05, 1000] |
| material (CVM)   | 194 × 361 ×<br>452 | 1500, 1500, 250                    | x[28.5,705]k<br>y[3543,4083]k<br>z[−45000,+3250] | rho, mu, lambda                                     |

| file                      | dims (z,y,x)               | $\Delta x, \Delta y, \Delta z$ [m] | extent [m]                                      | members                                              |
|---------------------------|----------------------------|------------------------------------|-------------------------------------------------|------------------------------------------------------|
| stress (Anders.<br>k=1.8) | $93 \times 281 \times 434$ | 1000, 1000, 250                    | x[195,628]k<br>y[3685,3965]k<br>z[−20000,+3000] | $s_{xx}..s_{xz}; s_{yz}$<br>$\equiv s_{xz} \equiv 0$ |

CASE2 friction profile ( $b = 0.019$  constant, slope  $m = 8.0e-5$  /°C):

$a - b = -0.004$  for  $T \leq 300$  °C (a = 0.015; VW everywhere  
 $\hookrightarrow$  shallow)  
 $a - b = -0.004 + m \cdot (T - 300)$  for  $T \geq 300$  °C (zero-crossing exactly at  
 $\hookrightarrow 350$  °C)  
 $V_w = 0.05$  m/s for  $T \leq 350$  °C  
 $V_w = 0.05 + (1000 - 0.05) \cdot (T - 350) / 50$  for  $350 < T < 400$  °C  
 $V_w = 1000$  m/s for  $T \geq 400$  °C

CASE1 (the sibling deck) adds a shallow VS cap at the 100 °C isotherm; **CASE2 has no shallow VS zone** — the shallow fault keeps  $V_w = 0.05$  and can weaken dynamically. At the hypocentre column: 100 °C at 2.93 km, 350 °C at 13.19 km depth.  $T(\text{hypocentre}) = 277.8$  °C  $\Rightarrow a = 0.015$ ,  $V_w = 0.05$ .

### The mesh is already here, and it is the same mesh

```
$ shasum -a 256 .../meshing_deep19km/results/safv4_deep19km_sub500m_flattop_nwtrim_f
  ↪ ixed_opt.puml.h5
      ~/Downloads/seis0l_quakeworx/.../mesh_preferred.puml.h5
f08aef34... (identical)
```

The PUML the SeisSol job read was produced by `msh_to_puml.py` from

`safv4_deep19km_sub500m_flattop_nwtrim_fixed_opt_safstags.msh`

whose default map is 101 fault  $\rightarrow$  3, 102 top  $\rightarrow$  1, 103 bottom  $\rightarrow$  5, 104 sides  $\rightarrow$  5. That `.msh` is Gmsh **v2.2 ASCII** (MFEM's only supported flavour), 250,588 nodes, 1,543,222 elements (1,241,419 tets plus 301,803 boundary triangles), physical tags 101/102/103/104 + volume 1.

`seas_spatial_dyn_driver` with **no** [boundary] block falls back to exactly `fault_attr=101`, `natural={102}`, `absorbing={103,104}` (`spatial_dyn_driver.cpp:1070`). So the mesh channel is **zero work**.

For scale: the MFEM SAFS jobs already in production run `safv4_deep_500m_opt_safstags.msh` — **314,284 nodes / 1,846,898 elements**, i.e. *larger* than the deep19km mesh — on  $2 \text{ nodes} \times 50 \text{ ranks}$ .

### What MFEM already implements

- `interior_flux="matrix" + mixed_flux="adjacent" + time_integrator="rk4"|"rk45"` is the intended heterogeneous-CVM mixed-flux mode. `BimaterialWaveOperator::SetMixedFluxMode` (`bimaterial_wave_operator.inl:1170`) explicitly *lifts* the old R-003 matrix×mixed-flux exclusion. Central flux is non-dissipative  $\Rightarrow$  **unstable under ADER**; both `ComputeMaxDt` overloads and a driver-level guard (`MatrixMixedFluxUnderAder`) abort on that combination.

The comment at `wave_operator.hpp:335` claiming “mixed flux is scalar-only” is **stale**.

- **FL=103 equivalent:** `state_evolution = "slip_law_strong_rate_weakening"` (alias `"slip_law_srw"`), `slip_law_srw_psi.hpp: f_ss(V) = f_w + (f_LV - f_w)/(1+(V/V_w)8)1/8`, `f_LV = max(0, f_0 - (b-a)·ln(V/V_0))`, `ψss = a·ln((2V_0/V)·sinh(fss/a))`. Per-QP `a` and per-QP `V_w` are threaded through `Rate_SRW / SteadyState_SRW / PsiSS_SRW`.
- **Compact-bell nucleation:** `[nucleation] kind = "gradual overstress_compact_circular"` → `CompactBellFactor(r,R) = exp(r2/(r2-R2))`, strike-only (`tau2_nuc`), SCEC smooth-Step temporal ramp over `[0, Tnuc_s]` — the same function `SeisSol` uses for `t_0`. Parser at `spatial_friction.cpp:1363`, driver dispatch at `spatial_dyn_driver.cpp:1911`. **Both CLAUDE.md (§ *Spatial driver nucleation mechanism*) and `spatial_friction_config_schema.md` claim the spatial driver has only `gradual overstress`, and that rate-state ignores `[nucleation]`. Both statements are stale** — `tpv104/configs/tpv104_spatial.toml` is `law="rate_state" + kind="gradual overstress_compact_circular"` and is a validated benchmark. Fix both docs as part of this work.
- **Stress sidecar:** `[stress] kind="sidecar_hdf5"` → `StressField3D (stress_field_3d.hpp)`, six `DataField3D` readers over a `data_projection_v1` HDF5 (`grid/{x,y,z}` float64 ascending, `fields/sigma_{xx,yy,zz,xy,yz,xz}` shaped `(Nx,Ny,Nz)`, `attrs schema_version, crs="EPSG:32611", z_positive="elevation", per-dataset min_value/max_value`). **Strict pass-through — compression POSITIVE.** The  $\sigma_{xy}$  negation at `spatial_dyn_driver.cpp:1716` is in the `ConstantTensor` branch **only**; the sidecar branch does not touch signs.
- **Material sidecar:** `[velocity]` bundle reads `fields/{Vp, Vs, density}` from the same schema; `far_field_clamp = true` ⇒ `00BPolicy::Clamp`, the ASAGI nearest-edge hold.  $\mu = \rho V_s^2$ ,  $\lambda = \rho V_p^2 - 2\mu$ . Evaluated per quadrature point (`Mode::Coefficient`), no P1 smearing.
- **Converters already written:**
  - `seisolv_quakeworx/toolbox/on_fault_stress_projection_csm/csm_stress_nc_to_mfem_hdf5.py` — ASAGI stress .nc → schema-v1 HDF5, applies the **global — 1 sign flip** on all six components and already nudges `max_value` for identically-zero components (which `s_yz/s_xz` are here).
  - `safs/project_7.0_preferred/velocity/code/build_pref_velocity_sidecar.py` — `safs_material_cvm.nc {rho,mu,lambda}` → schema-v1 HDF5 `{Vp,Vs,density}`.
- **Checkpoint/restart,** ParaView fault output (7 fields: `slip_{dip,strike}`, `slip_rate_{dip,strike}`, `traction_{dip,strike}`, `state_variable`), free-surface slice.

### Sign / frame conventions (established, audited 2026-07-07)

|        | MFEM                        | SeisSol                     | handoff                                        |
|--------|-----------------------------|-----------------------------|------------------------------------------------|
| Frame  | UTM 11N, x=E y=N<br>z=Up    | same                        | 1:1                                            |
| Stress | compression <b>positive</b> | compression <b>negative</b> | global $\times (-1)$ at the<br>converter, once |

|                     | MFEM                                                              | SeisSol                              | handoff                                                                                                |
|---------------------|-------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| Fault tangent frame | t1 = dip, t2 = strike<br>(FaultBasis/Tandem)                      | 1 = strike, 2 = dip                  | tau_strike_MFEM $\equiv$<br>Ts_SeisSol exactly<br>(two flips cancel);<br>tau_dip_MFEM =<br>-Td_SeisSol |
| ref_normal          | [0, -1, 0] (default)                                              | XRef=(0, -1, 0),<br>refPointMethod=1 | identical                                                                                              |
| $\sigma_n$          | > 0 compression                                                   | min( $\sigma_n$ , 0)                 | —                                                                                                      |
| Pore pressure       | $\sigma_{n\_eff} = \sigma_n -$<br>P_p, applied<br>unconditionally | —                                    | nc is <b>already effective</b><br>$\Rightarrow$ <b>P_p_pa = 0.0</b> (else<br>double-subtract)          |

The Andersonian nc's own global attribute states it: `convention = "SeisSol compression-NEGATIVE; effective (hydrostatic P_p removed)"`.

### 1.3 Gap table

| ID | Gap                                                                                                                                                                                                                                                                                                             | Severity       | Phase               |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|
| G1 | No 3-D field ingestion for rs_a / rs_srW. Resolver seeds a,b from scalars or a 1-D depth CSV; V_w from scalars, box rules, or a boxcar taper. <code>spatial_friction.cpp:2031</code> .                                                                                                                          | <b>Blocker</b> | 3                   |
| G2 | MFEM_VERIFY( <code>f_w_default &gt; 0.0 &amp;&amp; f_w_default &lt; 1.0</code> ) at <code>spatial_friction.cpp:713</code> rejects <code>f_w = 0</code> .                                                                                                                                                        | <b>Blocker</b> | 2                   |
| G3 | <code>velocity_safs.h5</code> was built from the v3.0.0 CVM nc (297×196×194); the deck ships the statewide nc (452×361×194). <code>build_pref_velocity_sidecar.py</code> also hard-codes a path under the old <code>seisolver/quakeworx/</code> location, <b>which no longer exists</b> (relocated 2026-06-26). | Data           | 4                   |
| G4 | No MFEM stress sidecar for <code>safs_stress_andersonian_k1.8.nc</code> .                                                                                                                                                                                                                                       | Data           | 5                   |
| G5 | No on-fault pickpoint / off-fault receiver time-series writer for SAFS. Station writers are hard-coded SCEC grids keyed on five fixed <code>[problem].tag</code> values. The deck ships 10 pickpoints + 10 receivers.                                                                                           | Feature        | 7<br>(opt.)         |
| G6 | No energy output (SeisSol <code>EnergyOutput=1</code> ).                                                                                                                                                                                                                                                        | Cosmetic       | not<br>planned      |
| G7 | MFEM has <b>no LTS</b> . SeisSol's clustered LTS gave a 30.5× speedup over GTS on this mesh.                                                                                                                                                                                                                    | Accept         | 1<br>(mea-<br>sure) |
| G8 | MFEM <code>[mesh].order = 1</code> (p1) vs SeisSol order 4 (p3). Not the same accuracy per DOF.                                                                                                                                                                                                                 | Decision       | 6                   |

| ID  | Gap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Severity                            | Phase |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------|
| G9  | <code>sigma_n_strength_floor_pa = 10e6</code> in every MFEM SAFS config. The Andersonian field has $\min \sigma_n = 2.15$ MPa on this fault, so <b>the floor is active</b> on shallow facets and makes them stronger than SeisSol. No SeisSol analog.                                                                                                                                                                                                                                         | Decision                            | 6     |
| G10 | Compact-bell radius: MFEM measures <b>r in the fault plane</b> (projection onto the per-DOF dip/strike basis, <code>spatial_nucleation.hpp:231</code> ); SeisSol's Lua uses the full 3-D Euclidean distance. On a curved fault the two differ by the normal offset; MFEM's $r \leq$ SeisSol's $r$ , so MFEM's bump is marginally wider.                                                                                                                                                       | Quantify                            | 6     |
| G11 | <code>spatial_dyn_driver.cpp:1418</code> warns: <code>rate_state</code> at $np > 1$ uses <b>end-of-step</b> $\psi$ on rank-seam fault QPs (1st order). Fires on every production SAFS run today.                                                                                                                                                                                                                                                                                              | Known                               | 8     |
| G12 | <i>(found during implementation)</i> <code>io/data_field_3d.cpp</code> holds each field's <b>entire</b> array on <b>every rank</b> ( <code>data_.assign(Nx*Ny*Nz,0)</code> + full H5Dread). <code>v3_4_1</code> 's three sidecars total <b>1329 MB/rank</b> ( <code>v3.0.0</code> : 779 MB) because the statewide CVM cube replaced the smaller one and friction is new. The stock $2N \times 50r$ / <code>--mem=100000M</code> sizing gives 2000 MB/rank — the sidecars alone are 66% of it. | <b>Blocker</b><br><b>(run-time)</b> | 8     |
| G13 | <i>(found during implementation)</i> <code>SEAS_HEADERS</code> omitted <code>spatial/code/*.hpp</code> , and this Makefile has no auto-dependency generation. Adding a member to <code>RateStateBlock</code> therefore did <b>not</b> rebuild <code>nucleation_factory.o</code> / <code>fault_face_flux.o</code> on an incremental make, silently linking mixed-ABI objects (observed: <code>seas_test_nucleation_factory</code> segfaulting).                                                | <b>Blocker</b><br><b>(build)</b>    | 3     |

## 1.4 Constraints

1. **Byte-exact regression contract.** TPV102 / TPV104 / TPV205 / BP5 outputs must not change. Every addition below defaults to “absent  $\Rightarrow$  old behaviour”. `make test` must stay green.
2. **Do not edit** the files `miniapps/seas/CLAUDE.md` lists as requiring extreme care:

|                                             |                                           |
|---------------------------------------------|-------------------------------------------|
| <code>friction/dieterich_ruina.hpp</code>   | <code>friction/state_evolution.hpp</code> |
| <code>domain/elasticity_operator.hpp</code> | <code>fault/fault_basis.hpp</code>        |
| <code>fault/rate_state_fault.hpp</code>     | <code>solver/seas_operator.hpp</code>     |
| <code>solver/time_stepper.hpp</code>        | <code>config/bp5_params.hpp</code>        |

Nothing in this plan needs to.
3. **No NetCDF in C++.** The driver links HDF5 only (`data_field_3d.cpp` uses the raw HDF5 C API). All `.nc`  $\rightarrow$  `.h5` conversion stays in offline Python. Do not add a `netcdf-c` dependency.
4. **Gmsh v2.2 ASCII only** for `[mesh].path`.
5. **Frontera/Expanse runs need explicit user approval** before submission.

6. Mixed flux  $\Rightarrow$  RK. `mixed_flux != "none"` under ADER aborts by design; do not “fix” that.
- 

## 1.5 Phase 1 – Baseline measurement (no code)

### Goal

Know the actual time-step and per-step cost on the deep19km mesh before committing to a wall-time budget.

### Steps

1. Build the driver locally (`conda activate mfem-dev; make seas_spatial_dyn_driver`).
2. Author a throwaway config that points at the deep19km .msh with `--no-sidecar-material` (constant material), `interior_flux="scalar"`, `mixed_flux="none"`, `time_integrator="ader"`, and run `mpirun -np 8 ./seas_spatial_dyn_driver --config <tmp>.toml --dry-run --print-derived`. Record `h_min`, `dt_cfl`.
3. Repeat with `--time-integrator rk4 --mixed-flux adjacent` (still constant material, scalar flux) and record `dt_cfl`.

### Expected numbers (derive, then check)

$h$  is the **inscribed diameter**  $6 \cdot \text{Vol} / A_{\text{total}}$  (`wave_operator.inl:96`) — the same  $2 \cdot r_{\text{insphere}}$  SeisSol uses. SeisSol reported `dt_min = 5.13722e-5 s` at order 4 ( $N=3$ ), CFL 0.5, from `dt = cfl \cdot 2r / ((2N+1) \cdot c_p)`, hence `min_e(h_e/c_p,e) = 7 \cdot 5.13722e-5 / 0.5 = 7.192e-4 s`.

MFEM: `dt = MixedFluxCflFactor \cdot (cfl \cdot RkCflFactor) \cdot min_e(h_e/c_p,e)` where `RkCflFactor = 3/(2p+1) = 1.0` at  $p=1$  and `MixedFluxCflFactor(Adjacent, rk_aware) = 0.6` (`wave_operator.hpp:870`; `spatial_friction.hpp:711`; `spatial_dyn_driver.cpp:2221`):

`dt(p=1, rk4, adjacent) = 0.6 \cdot 0.5 \cdot 7.192e-4 = 2.16e-4 s \rightarrow \sim 695,000 \text{ steps} \times 4`  
 $\hookrightarrow$  RHS for 150 s

`dt(p=1, ader, upwind, cfl_dg_safety=3) = (0.5/9) \cdot 7.192e-4 = 4.00e-5 s \rightarrow \sim 3.75M`  
 $\hookrightarrow$  steps

`dt(p=1, ader, upwind, cfl_dg_safety=1) = (0.5/3) \cdot 7.192e-4 = 1.20e-4 s \rightarrow \sim 1.25M`  
 $\hookrightarrow$  steps

Note `--cfl-dg-safety` is **inert on the RK path** — it only scales `CflSafetyFactor`, which the ADER branch uses. The mixed-flux+RK production target gets no benefit from Lever A.

For calibration: SeisSol reached only  $t \approx 98$  s of 150 s in 300 min on 160 ranks (job hit the Tapis time limit), *with* a  $30.5\times$  LTS speedup. Expect MFEM GTS at  $p1$  to need a **checkpoint/restart chain**, not a single 2 h job.

### Acceptance

`dt_cfl` printed and within  $2\times$  of the estimate above. If it is wildly smaller, a sliver tet is driving `h_min` and that must be investigated before anything else (see `project_flattop_badtet_fix / QUALITY_REPORT_deep19km.md`, which reports `eta_min PASS`).

### RESULT (2026-07-09, np=1 --dry-run --print-derived, production parity config)

[time]     `dt_cfl = 0.000215799 s`                      predicted `2.16e-4 s`     -- MATCHES



```
[derived] nsteps    = 695,093                predicted ~695,000    -- MATCHES
[derived] mesh h_min = 70.5003 m
[mixed-flux] mode = adjacent, central-flux faces (rank-summed) = 708,810
```

dt\_cfl lands on the prediction to 3 significant figures. That is a genuine cross-check, not a coincidence: the estimate was derived from SeisSol's own reported  $dt_{\min} = 5.13722e-5$  s at order 4, inverted to  $\min_e(h_e/c_p, e) = 7.192e-4$  s. MFEM independently measures the same quantity from the same mesh through a different code path (6·Vol/A<sub>total</sub> vs SeisSol's inscribed-sphere diameter) and agrees. No sliver tet is driving h<sub>min</sub>.

Setup cost at np=1 was 851 s wall, 15.7 GB peak RSS (serial mesh + all three sidecars). Per-rank at np=100 the mesh share collapses but the 1329 MB of sidecars does **not** — see G12.

## 1.6 Phase 2 – Allow $f_w = 0$ (G2)

### Goal

RS\_muW = 0.0 (zero residual friction — the rate-state analogue of LSW  $\mu_d = 0$ ) must parse.

### Why it is safe

$f_w$  enters only additively:  $f_{ss} = f_w + (f_{LV} - f_w)/(1+(V/V_w)^8)^{1/8}$ . There is no division by  $f_w$  and no  $\log(f_w)$ .  $\psi_{ss} = a \cdot \ln((2V_0/V) \cdot \sinh(f_{ss}/a))$  needs  $f_{ss} > 0$ ; with  $f_w = 0$ ,  $f_{ss} = f_{LV}/(1+(V/V_w)^8)^{1/8}$  and  $f_{LV} = \max(0, f_0 - (b-a) \cdot \ln(V/V_0))$  only reaches 0 at  $V = V_0 \cdot \exp(f_0/(b-a)) = 1e-6 \cdot e^{150}$ , which is unreachable.  $f_{ss} > 0$  strictly  $\Rightarrow \psi_{ss}$  finite. SeisSol runs this deck with RS\_muW = 0 on v1.1.3. The deck header flags it as an “energy-max idealization below Lachenbruch & Sass  $\mu_d \approx 0.10$ ”.

### Files to change

- spatial\_friction.cpp:713 — relax to  $f_w_{\text{default}} \geq 0.0$  &&  $f_w_{\text{default}} < 1.0$ ; update the R-004 comment to record *why* 0 is admissible and that  $V_w > 0$  is the load-bearing guard (never  $V_w = 0$  under FL=103).

### Files to create

- seas\_test\_slip\_law\_srw\_fw\_zero/ (or extend seas\_test\_slip\_law\_srw\_psi): assert PsiSS\_SRW( $V, V_w, a$ ) is finite and monotone for  $f_w = 0$  across  $V \in [1e-12, 10]$  m/s with  $a=0.015$ ,  $b=0.019$ ,  $f_0=0.6$ ,  $V_0=1e-6$ ,  $V_w=0.05$ ; assert the parser accepts  $f_w_{\text{default}} = 0.0$  and still rejects  $-0.1$  and  $1.0$ .

### Acceptance

- make test green.
- ParseSpatialFrictionConfigString with  $f_w_{\text{default}} = 0.0$  + state\_evolution = "slip\_law\_srw" returns without abort.

## 1.7 Phase 3 – 3-D friction-field sidecar (G1) – the only real feature work

## Goal

Let [friction.rate\_state] seed per-DOF  $a$  and  $V_w$  from a data\_projection\_v1 HDF5, exactly as [stress] and [velocity] already do. This is what carries the CTM temperature zoning.

## Design

Mirror the stress/velocity pattern. Do **not** invent a new file format; reuse DataField3D (trilinear, 00BPolicy::Clamp = ASAGI edge-hold).

**Why baked  $a/V_w$  and not a temperature field + inline  $a(T)$  formula:** SeisSol itself reads baked  $rs_a/rs_{srW}$  through ASAGI. Reading the same two fields means the two codes sample *identical* friction, and the  $a(T)/V_w(T)$  piecewise logic stays in one place (build\_friction\_nc\_thermal.py) instead of being duplicated in TOML.

**1.7.2.1 Python: safs/project\_7.0\_preferred/friction/code/build\_pref\_friction\_sidecar.py (new) Decision D5 (user, 2026-07-08): MFEM never reads .nc. The friction sidecar is the same file type as the velocity sidecar** — a data\_projection\_v1 HDF5, written by the **same shared helper** safs/project\_7.0\_alternative/velocity/code/sidecar.py::write\_sidecar that build\_pref\_velocity\_sidecar.py already uses. Do **not** hand-roll an HDF5 writer (the stress converter has its own inline copy; that is technical debt, not a template).

So this script is a sibling of build\_pref\_velocity\_sidecar.py, not of csm\_stress\_nc\_to\_mfem\_hdf5.py:

```
# in : safs_friction_thermal_case2.nc    compound data{rs_a, rs_srW}, dims (z,y,x), f4
# out: ../friction/results/thermal_case2_mfem/friction_safs.h5
sys.path.insert(0, f"{ALT}/velocity/code")
from sidecar import write_sidecar

data = netCDF4.Dataset(NC).variables["data"][:]          # (z, y, x) compound
rs_a  = np.transpose(np.asarray(data["rs_a"],    np.float64), (2, 1, 0))  # -> (x,
    ↪ y, z)
rs_srW = np.transpose(np.asarray(data["rs_srW"], np.float64), (2, 1, 0))

write_sidecar(
    OUT, x, y, z,
    fields      = {"rs_a": rs_a, "rs_srW": rs_srW},
    attrs       = {"schema_version": "data_projection_v1", "crs": "EPSG:32611",
                  "units": "m", "z_positive": "elevation",
                  "source": "safs_friction_thermal_case2.nc (CTM shinevar2024, CASE2
    ↪ "
                  "a(T)/V_w(T)) -> build_pref_friction_sidecar.py",
                  "mesh_tag": "safv4_deep19km_sub500m_flattop_nwtrim_fixed_opt"},
    field_bounds = {"rs_a": (0.010, 0.050, "1"),
                  "rs_srW": (0.05, 1000.0, "m/s")},
)
```

write\_sidecar already enforces: strictly-monotone axes, no NaN, shape (Nx,Ny,Nz), and every cell

inside `field_bounds` — i.e. it gives the build script's G2 gate ( $rs\_a > 0$ ,  $rs\_srW \in [0.05, 1000]$ ) for free and raises rather than warns.

Add one guard the helper cannot know about: `assert a(hypocentre) == 0.015` and `V_w(hypocentre) == 0.05` to `1e-9` by trilinear sampling before writing (the deck's own G4 gate).

Resulting layout, byte-comparable to `velocity_safs.h5`:

```
grid/x (263,) f8  grid/y (194,) f8  grid/z (107,) f8      # ascending, equidistant
fields/rs_a  (263,194,107) f8  attrs units="1",  min_value, max_value
fields/rs_srW (263,194,107) f8  attrs units="m/s", min_value, max_value
```

### 1.7.2.2 C++: `schema spatial_friction.hpp`:

```
/// [friction.rate_state.sidecar] — per-DOF a / V_w (and optionally b, Dc) from a
/// data_projection_v1 HDF5. Absent  $\Rightarrow$  enabled=false  $\Rightarrow$  resolver behaviour
 $\hookrightarrow$  byte-identical.
struct FrictionSidecarSpec
{
    bool          enabled          = false;
    std::string path;              // required when enabled
    std::string a_field            = "rs_a";
    std::string V_w_field          = "rs_srW";
    std::string b_field;           // ""  $\Rightarrow$  keep b_default
    std::string Dc_field;          // ""  $\Rightarrow$  keep Dc_default
    bool          far_field_clamp = true;  // OOBPolicy::Clamp (ASAGI edge hold)
};
```

Add `FrictionSidecarSpec sidecar;` to `RateStateBlock`.

Loaded fields (owned by the driver, handed to the resolver):

```
struct RateStateSidecarFields
{
    std::unique_ptr<DataField3D> a, V_w, b, Dc;  // b/Dc may be null
};
```

`SpatialFrictionResolver` gains an optional ctor argument `std::shared_ptr<const RateStateSidecarFields>`; the default ctor leaves it null, so every existing call site (and every TPV/BP5 test) is byte-unchanged.

### 1.7.2.3 C++: `parser (spatial_friction.cpp, in parse_rate_state)`

- Read `[friction.rate_state.sidecar]`; path required and non-empty when the table is present.
- `MFEM_VERIFY(!(sidecar.enabled && depth_profile.enabled))` — both seed a/b; a config that sets both is ambiguous and must abort with a message naming both keys.
- Keep the existing `a_default < b_default` guard satisfied by the config (`0.015 < 0.019`); `a_default` becomes a pure fallback.

**1.7.2.4 C++: resolver (resolve\_rs\_impl, around spatial\_friction.cpp:2031)** Insert **after** the depth-profile/scalar seed and **before** the spatial-rule loop, so box / boxcar\_taper rules can still override a sidecar value:

```
if (sidecar_) // null ⇒ untouched legacy path
{
    a_i = sidecar_>a ->Evaluate(x, y, z);
    V_w_i = sidecar_>V_w->Evaluate(x, y, z);
    if (sidecar_>b) { b_i = sidecar_>b ->Evaluate(x, y, z); }
    if (sidecar_>Dc) { Dc_i = sidecar_>Dc->Evaluate(x, y, z); }
}
```

The existing  $V_w_i > 0$  guard (:2166) and the per-DOF  $a > b$  allowance (R-011, :2153) already cover the VS zone ( $a$  up to  $0.0332 > b = 0.019$ ).

#### 1.7.2.5 C++: driver (spatial\_dyn\_driver.cpp)

- Construct RateStateSidecarFields right before ResolveRateState (after the mesh + material exist).
- 00BPolicy = Clamp when far\_field\_clamp, else Abort.
- Extend --print-derived to report, over fault DOFs: min/max/mean of  $a$  and  $V_w$ , the count of DOFs with  $a - b < 0$  (the VW area), and the values at [hypocenter]. This is the offline↔online cross-check against the SeisSol build script's own G4 gate.

#### Files to create

- safs/project\_7.0\_preferred/friction/code/build\_pref\_friction\_sidecar.py
- seas\_test\_spatial\_friction\_sidecar/ — three tests:
  1. **Round-trip**: write a synthetic  $3 \times 3 \times 3$  sidecar with a known linear  $a(x,y,z)$ ; resolve at 8 corners and the centre; assert trilinear values to  $1e-12$ .
  2. **Clamp**: query outside the hull with far\_field\_clamp = true ⇒ edge value; with false ⇒ abort.
  3. **Rule precedence**: sidecar + a box rule setting  $V_w \Rightarrow$  rule wins inside the box, sidecar outside.
- Parser tests: sidecar + depth\_profile together ⇒ abort; missing path ⇒ abort.

#### Files to change

- spatial\_friction.hpp (schema + resolver ctor)
- spatial\_friction.cpp (parse + resolve)
- spatial\_dyn\_driver.cpp (load + --print-derived)
- safs/project\_7.0\_alternative/document/spatial\_friction\_config\_schema.md (document the new block; also fix the stale “rate-state ignores [nucleation]” claim at lines 256-258 / 376-377)
- miniapps/seas/CLAUDE.md (fix the stale “single nucleation kind” paragraph)

## Acceptance

- make test green; seas\_test\_spatial\_friction\_resolver byte-unchanged (no sidecar  $\Rightarrow$  no behaviour change).
- `--print-derived` on the real sidecar reports  $a(\text{hypo}) = 0.015$ ,  $V_w(\text{hypo}) = 0.05$ ,  $\min a = 0.015$ ,  $\max a \approx 0.0332$ ,  $\min V_w = 0.05$ ,  $\max V_w = 1000$ .
- Fault-DOF VW fraction is consistent with “350 °C isotherm at  $\sim 13.2$  km at the hypocentre column”.

## RESULT (2026-07-09, np=1 --dry-run --print-derived, production parity config)

```
[print-derived] friction sidecar (a, V_w)
  fault DOFs (global) : 517779
  a      min/mean/max : 0.015 / 0.0177678 / 0.0298597
  V_w    min/mean/max : 0.05 / 225.437 / 1000 m/s
  velocity-weakening : 371373 / 517779 DOFs with a - b < 0 (71.72 %)
  sidecar at [nucleation.gradual overstress compact circular] centre (609063,
  ↪ 3.70953e+06, -10000):
    a = 0.015          <- the CASE2 VW-plateau anchor
    V_w = 0.05 m/s     <- ditto
```

All gates met, with one refinement to the wording above:  $\max a = 0.0299$  **on the fault**, not 0.0332. 0.0332 is the maximum over the whole CTM *grid*; the fault bottom ( $-19.3$  km) never reaches the grid’s hottest cells. The two numbers are consistent, and the offline builder’s own G2 gate still checks the grid-wide 0.0332263.

The VW fraction (71.7 %) matches the geometry: the 350 °C isotherm sits at  $\approx 13.2$  km at the hypocentre column and the fault runs to  $-19.33$  km, so  $\approx 13.2/19.33 \approx 68$  % of a uniformly-sampled fault should be VW; 71.7 % reflects the shallower isotherm away from the hypocentre column.

seas\_test\_spatial\_friction\_resolver stayed at 127/127 (byte-unchanged), and 24 affected test binaries pass from a clean build.

---

## 1.8 Phase 4 – Rebuild the CVM material sidecar (G3)

### Goal

velocity\_safs.h5 sampled from the **same** safs\_material\_cvm.nc the SeisSol job read.

### Files to change

- safs/project\_7.0\_preferred/velocity/code/build\_pref\_velocity\_sidecar.py
  - Replace the hard-coded source path

```
NC = f"{ALT}"
↪ /seisol_quakeworx/safs_seisol_v3_0_0_LSW_PREFERRED/safs_material_cvm.nc"
```

(this path no longer exists — seisol\_quakeworx moved to safs/seisol\_quakeworx/ on 2026-06-26) with `--in` / `--out` argparse options. Keep the old defaults out; make both required.

## Output

project\_7.0\_preferred/velocity/results/multiscale\_statewise\_cvm\_v3\_4\_1/velocity\_safs.h5  
(new directory so the v3.0.0 sidecar is not clobbered; both are gitignored, scp to Expanse).

Source nc: safs\_material\_cvm.nc from the deck directory ( $452 \times 361 \times 194$ ; the existing h5 is  $297 \times 196 \times 194$  — a **different, smaller** grid).

## Acceptance

- $v_p = \sqrt{(\lambda + 2\mu)/\rho}$ ,  $v_s = \sqrt{\mu/\rho}$ , density =  $\rho$ , all finite,  $\rho > 0$ ,  $\mu \geq 0$ .
  - Grid attrs: crs="EPSG:32611", z\_positive="elevation", axes ascending.
  - Mesh bbox x[253447,711231] y[3532686,3956757] z[-39329,+26] pokes outside the CVM hull only at the far-field absorbing corners (x high  $\leq 6.2$  km, y low  $\leq 10.3$  km)  $\Rightarrow$  **far\_field\_clamp = true is mandatory**, exactly as SeisSol relies on ASAGI's edge clamp + the yaml !ConstantMap.
- 

## 1.9 Phase 5 – Convert the Andersonian stress field (G4)

### Goal

Andersonian effective stress  $\rightarrow$  schema-v1 HDF5, compression-**positive**, effective Pa.

### STRESS REVISION (2026-07-09, user directive): **k=1.5 + freeze500m**

The original port converted the deck's safs\_stress\_andersonian\_k1.8.nc. The stress source was then **changed to the SeisSol-verified v3\_5\_3 field safs\_stress\_andersonian\_k1.5\_freeze500m.nc** so that shallow free-surface facets cannot free-slip at near-zero normal stress (freeze500m: every node shallower than 500 m carries its column's stress at 500 m depth). This deviates from the deck's k=1.8 — it is the same Andersonian construction at a lower differential-stress ratio (k=1.5) with the frozen shallow cap.

```
python3 safs/seis0l_quakeworx/toolbox/on_fault_stress_projection_csm/csm_stress_nc_
↪ to_mfem_hdf5.py \
--pad-m 3000 \
--in safs/seis0l_quakeworx/v3_under_construction/toolbox/combined_workflow/output_
↪ s/safs_stress_andersonian_k1.5_freeze500m.nc \
--out safs/project_7.0_preferred/stress/results/andersonian_k1p5_freeze500m_mfem/s_
↪ afs_stress_andersonian_k1.5_freeze500m_mfem.h5 \
--hypo 609062.8722 3709528.1324 -10000
```

The script needs no modification: members are s\_xx...s\_xz, the min<max nudge covers s\_yz  $\equiv$  s\_xz  $\equiv 0$ .

### Cautions

- **--pad-m 3000 is REQUIRED.** The freeze500m nc caps at z=0, but the deep19km flat-top fault daylights its trace to  $z \approx +25.8$  m. pad\_grid edge-replicates the z=0 (frozen) slice up to z=+3000, so the daylighting facets stay inside the strict-Abort stress hull and read the frozen top stress. (The old k=1.8 nc already extended to +3000; this reproduces that.) Without it, ApplyCsmStressSidecar (00BPolicy::Abort) aborts on the trace facets.

- **[pore\_pressure] P\_p\_pa = 0.0.** The nc is already effective; a non-zero P\_p\_pa double-subtracts.
- The script's HYP0\_DEFAULT is the v3.0.0 hypocentre (diagnostic print only); pass --hypo with the v3.4.1 on-fault point.

### Acceptance (measured on the deep19km fault, 2026-07-09)

- Hypocentre facet (144 m from the centre,  $z = -9932$  m):  $\sigma_n = 167.26$  MPa,  $|\tau_0| = 21.35$  MPa,  $\mu_{app} = 0.128 < f_0$ .
- $\min \sigma_n$  over the fault = **5.39 MPa** (was 2.15 at  $k=1.8$ ) — freeze500m raised the shallow floor  $2.5\times$ ; 40 shallowest facets 5.4–8.5 MPa, no near-zero. 0 non-finite. 0 fault facets outside the padded hull.
- **Nucleation consequence (flagged):** the binding static excess  $(f_0 - \mu_{app}) \cdot \sigma_n$  ROSE to **79.0 MPa** (from 72.1 at  $k=1.8$ ), because lower  $k$  dropped  $\mu_{app}$  more than  $\sigma_n$ . The **75 MPa bump is now below it ( $\times 0.949$ )** → **will not nucleate**. The v3\_5\_3 decks that ran this exact stress used **85 MPa** ( $\times 1.076$ ). This is Gate-4 decision territory (Phase 8); the bump was NOT changed automatically.

## 1.10 Phase 6 – Production TOML

### File to create

safs/project\_7.0\_preferred/config/

spatial\_friction\_safs\_seisol\_v3\_4\_1\_RSSRW\_PREFERRED\_THERMAL\_CASE2.toml

[meta]

schema\_version = 1

law = "rate\_state"

description = "MFEM port of

↪ seisol\_quakeworx/safs\_seisol\_v3\_4\_1\_RSSRW\_PREFERRED\_THERMAL\_CASE2 (FL=103, CTM

↪ temperature-zoned rs\_a/rs\_srW via sidecar,  $f_w=0$ , Andersonian  $k=1.8$  effective

↪ stress, statewide CVM) with MIXED FLUX (adjacent) + RK. 'Do what SeisSol did, but

↪ with mixed flux.'"

[material\_constant\_fallback]

# only reachable via --no-sidecar-material

lambda = 32.0e9

mu = 32.0e9

rho = 2670.0

[pore\_pressure]

P\_p\_pa = 0.0

# nc is EFFECTIVE (hydrostatic Pp already

↪ removed)

P\_p\_grad\_pa\_per\_m = 0.0

min\_sigma\_n\_pa = 1.0e6

# inert: min sigma\_n on this fault is 2.15 MPa

[mesh]

```

path = "safs/project_7.0_preferred/meshing_deep19km/results/safv4_deep19km_sub500m_
↳ flattop_nwtrim_fixed_opt_safstags.msh"
order = 1                                # DECISION D2 (SeisSol runs p3)

[velocity]
use_sidecar      = true
model            = "multiscale_statewise"
dataset_root     = "safs/project_7.0_preferred"
override_path    = "safs/project_7.0_preferred/velocity/results/multiscale_statewise_
↳ cvm_v3_4_1/velocity_safs.h5"
far_field_clamp = true                  # MANDATORY: mesh far field exceeds CVM hull

[material]
kind             = "sidecar_hdf5"        # required: interior_flux="matrix" forbids
↳ kind="constant"
sidecar_path     = "safs/project_7.0_preferred/velocity/results/multiscale_statewise_cvm_
↳ _v3_4_1/velocity_safs.h5"

[stress]
kind            = "sidecar_hdf5"
sidecar_path    = "safs/project_7.0_preferred/stress/results/andersonian_k1p8_mfem/safs_
↳ _stress_andersonian_k1.8_mfem.h5"

[numerics]
ader_order      = 2                    # ignored on the RK path
mixed_flux      = "adjacent"
interior_flux   = "matrix"            # heterogeneous CVM ⇒ bimaterial Riemann
time_integrator = "rk4"               # central flux is UNSTABLE under ADER
cfl             = 0.5                 # = SeisSol CFL
use_pml         = false

[time]
tfinal         = "150s"                # = SeisSol EndTime
t_initial      = 0.0
dt_initial     = "auto"
dt_max         = "0.1s"                # = SeisSol FixTimeStep

[output]
output_dir      = "output_safs_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2"
restart_prefix  = "cp"
paraview_volume = "off"
paraview_bulk   = "off"
paraview_fault  = "hdf5"
paraview_fault_dt = "4.0s"            # = SeisSol Elementwise printtimeinterval_sec
paraview_fault_zfp_tol = 1.0e-12

```



```

paraview_free_surface      = "vtu"
paraview_free_surface_dt  = "4.0s"          # = SeisSol SurfaceOutputInterval
max_snapshots              = 5000
checkpoint_every_steps     = 10000

[nucleation]
kind = "gradual overstress compact circular"

[nucleation.gradual overstress compact circular]
# = SeisSol Tnuc_s = 75e6 * exp(r^2/(r^2-R^2)), R=2000, smoothStep over [s_0,
  ↪ s_0+t_0] = [0,1] s.
# Centre = the on-fault point at EXACTLY 10 km depth closest to the deck-lineage
  ↪ geographic target.
center_x_m    = 609062.8722
center_y_m    = 3709528.1324
center_z_m    = -10000.0
radius_m      = 2000.0
delta_tau_pa  = 75.0e6
T_nuc_s       = "1.0s"

[friction]
sigma_n_strength_floor_pa = -1.0          # DECISION D1: -1 = disabled = SeisSol parity

[friction.rate_state]
state_evolution = "slip_law_strong_rate_weakening" # = SeisSol FL=103
f_w_default     = 0.0          # = RS_muW (requires Phase 2)
V_w_default     = 0.05         # FALLBACK ONLY (sidecar governs)
a_default       = 0.015        # FALLBACK ONLY (sidecar governs); must stay < b_default
b_default       = 0.019        # = RS_b (constant; v1.1.3 reads no spatial rs_b)
Dc_default      = 0.10         # = rs_sl0 (constant)
V_0_default     = 1.0e-6       # = RS_sr0
f_0_default     = 0.6          # = RS_f0
V_init_default  = 1.0e-12      # = RS_iniSlipRate1
sigma_n_default = 169.83e6     # FALLBACK ONLY (live sigma_n is per-DOF from the stress
  ↪ sidecar)
eta             = "auto"       # 0.5*sqrt(mu*rho) per DOF == SeisSol radiation damping

[friction.rate_state.sidecar]          # NEW (Phase 3) – same file type as
  ↪ [velocity]
path      = "safs/project_7.0_preferred/friction/results/thermal_case2_mfem/friction_saf
  ↪ s.h5"
a_field   = "rs_a"
V_w_field = "rs_srW"
far_field_clamp = true              # CTM hull covers the fault, not the far field

```

```
# No [boundary] block: the driver falls back to fault=101, natural={102},
  ↪ absorbing={103,104},
# which is exactly what the safstags mesh uses.
# No [fault_geometry] block: defaults ref_normal=[0,-1,0], up=[0,0,1] ==
  ↪ XRef/refPointMethod=1.
# No [friction.rate_state.depth_profile]: mutually exclusive with [.sidecar].
```

### Decisions – RESOLVED (user, 2026-07-08)

**D1 —  $\sigma_n$  strength floor pa: two arms.** Every existing MFEM SAFS RS config sets  $10.0e6$ . On this fault  $\min \sigma_n = 2.15$  MPa, so the floor is **active** on shallow facets and makes them stronger than SeisSol's. Author **both** configs:

| config                           | $\sigma_n$ strength floor pa | role                                                                                                   |
|----------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------|
| ..._THERMAL_CASE2.toml           | -1.0 (disabled)              | <b>parity arm — run this first</b><br>fallback arm, only if the parity<br>arm shows trace-cell runaway |
| ..._THERMAL_CASE2_snfloor10.toml | $10.0e6$                     |                                                                                                        |

The two files differ in **that one key and output\_dir** and nothing else. Runaway signature (from project\_pre\_k25\_case2\_runaway\_slip\_not\_mesh): isolated shallow cells at  $\sigma_n$  0.1–5 MPa sliding  $\sim 4$  m/s indefinitely, visible in the first few fault snapshots.  $\min\_sigma\_n\_pa = 1.0e6$  is a *different* knob (it clamps  $\sigma_n$  eff, not the strength) and is inert here — keep it in both.

**D2 — [mesh].order: p1 production + p2 convergence window.**  $order = 1$  for the 150 s run (matches every existing SAFS production run and the Phase-1 cost estimate). Separately run  $order = 2$  with  $t_{final} = "20s"$  as a convergence check: if the p2 rupture front and peak slip rate agree with p1 to within the snapshot cadence, p1 resolves the cohesive zone well enough; if not, the p1 result is quantitatively suspect and must be reported as such. Under mixed-flux+RK, p2's dt penalty is only  $RkCflFactor(2)/RkCflFactor(1) = 0.6$ , but DOFs rise  $\sim 2.5\times \Rightarrow \sim 4\times$  total cost. Add this as gate 6b in Phase 8.

**D3 — time\_integrator: rk4.** Fewer RHS evaluations per step than rk45 at the same fixed dt. The existing sbatch scripts pass `--time-integrator rk45`; the sbatch override wins, so set the default in the sbatch to rk4 for this case.

**D4 — tfinal: 150 s (= EndTime).** But the reference SeisSol run only reached  $t \approx 98$  s before its Tapis time limit, so **0–98 s is the entire comparison window** the reference data supports.

**D5 — friction sidecar file type: data\_projection\_v1 HDF5,** identical in kind to the velocity sidecar, written by the shared sidecar.py::write\_sidecar. MFEM reads no NetCDF anywhere. See Phase 3.

**D6 — receivers/pickpoints (Phase 7): deferred** to a follow-up. Phases 1–6 + 8 give a runnable, comparable simulation on fault snapshots and free-surface slices; the per-point .dat writers upgrade that to waveform misfit once there is a run worth comparing.

### 1.11 Phase 7 – SAFS receiver / pickpoint output (G5) – optional, scope decision

Without this, the only comparison surface is ParaView fault snapshots at 4 s cadence and free-surface slices. The deck's `safs-faultreceiver-*.dat` (10 on-fault, `printtimeinterval = 10` steps) and `safs-receiver-*.dat` (10 off-fault, `pickdt = 0.005 s = 100 Hz`) are the highest-signal comparison data and they already exist for the SeisSol side.

#### Design

Generalize the existing `wire_stations` block (`spatial_dyn_driver.cpp:3297`) rather than adding a fifth hard-coded SCEC grid:

- New TOML: `[stations] on_fault_file = "...", off_fault_file = "...", on_fault_dt, off_fault_dt.`
- Reuse the `.dat` format the deck ships (whitespace-separated `x y z`, one point per line — see `safs_pickpoints.dat` / `safs_receivers.dat`, both 10 lines).
- On-fault: nearest fault DOF per point (assert the residual is  $< 0.5 \cdot \text{median fault edge} = 200 \text{ m}$ ); write `t`, `slip_strike`, `slip_dip`, `slip_rate_strike`, `slip_rate_dip`, `traction_strike`, `traction_dip`, `sigma_n`, `psi`.
- Off-fault: MFEM point-location on the volume mesh; write `t`, `vx`, `vy`, `vz`. Receivers sit at `z = -1 m` precisely because SeisSol v1.1.3 drops `z = 0` points; MFEM has no such restriction, but keep `z = -1` so both codes sample the same location.

#### Recommendation

**Defer to a follow-up.** Phases 1–6 + 8 produce a runnable, comparable simulation. Phase 7 upgrades the comparison from qualitative (rupture front, slip map) to quantitative (waveform misfit). Do it once the first run has produced something worth comparing.

---

### 1.12 Phase 8 – Job scripts and validation

#### Files to create

```
jobs/safs/safs_expanse/  
  spatial_dyn_ratestates_v3_4_1_pref_thermal_case2_cvm_mixedflux_rk_deep19km_expanse_  
  ↪ 2N100r_compute_2hr_safs.sbatch  
  ..._2hr_restart_safs.sbatch          # the --restart sibling
```

Copy the existing `..._v3pref_vw002_cvm_mixedflux_rk_safv4deep..._2hr_safs.sbatch` verbatim and change only `SAFS_CONFIG`, `SAFS_MESH`, `SAFS_OUT`, and the job name. Keep the existing `mixed_flux != none && integrator == ader abort guard`.

**Promote the validated perf levers:** `--deriv-cache` is  $\leq 1e-12$ -parity and works on the RK path (`--face-cache` and `--shared-ck-recursion` are ADER-only; `--cfl-dg-safety` is inert on RK). Today's production scripts pass none of them.

Sidecars to scp to Expanse alongside the mesh (all gitignored, ~700 MB total): `velocity_safs.h5` (from Phase 4), `safs_stress_andersonian_k1.8_mfem.h5` (Phase 5), `safs_friction_thermal_case2_mfem.h5` (Phase 3).

## Validation gates, in order

1. **--dry-run --print-derived, np=1, local.** Confirms mesh reads, all three sidecars load, boundary attrs resolve to 101/102/103/104, `dt_cfl` matches Phase 1.
2. **--print-derived friction gate.**  $a(\text{hypo})=0.015$ ,  $V_w(\text{hypo})=0.05$ ,  $\min/\max a = 0.015/0.0332$ ,  $\min/\max V_w = 0.05/1000$ . Compare against `build_friction_nc_thermal.py`'s G2/G4 gates.
3. **Stress gate.**  $\sigma_n(\text{hypo facet}) = 169.83$  MPa;  $\min \sigma_n$  over fault DOFs  $\approx 2.15$  MPa;  $\max \mu_{\text{app}} \approx 0.2931 < f_0 \Rightarrow$  **no t=0 pre-slip.**
4. **Nucleation gate. — RESOLVED (2026-07-09): 85 MPa on the k=1.5 + freeze500m stress.**

History (k=1.8 stress): the deck's 75 MPa bump cleared the static excess by  $\times 1.040$  but MFEM's stricter  $f_{\text{ss}} \cdot \sigma_n$  criterion in `--print-derived` gave nucleation overshoot =  $-6.86$  MPa — it would not nucleate. (MFEM compares  $|\tau_{\text{pre}} + \Delta\tau|$  against the *steady-state* strength  $f_{\text{ss}} \cdot \sigma_n$ , not  $f_0 \cdot \sigma_n$ .)

Stress revised to **k=1.5 + freeze500m** (Phase 5), which raised the binding static excess to **79.0 MPa** (lower k dropped  $\mu_{\text{app}}$  more than  $\sigma_n$ ), so the 75 MPa bump fell to  $\times 0.949$ . Per the **user directive (2026-07-09)** the bump was **raised to 85 MPa** in both configs — matching the SeisSol-run v3\_5\_3 PREFERRED CASE2 deck for this exact stress:

| criterion (k=1.5 hypocentre facet)                                             | 75 MPa                 | 85 MPa                  |
|--------------------------------------------------------------------------------|------------------------|-------------------------|
| RS barrier $S_E = 69.32$ MPa                                                   | $\times 1.082$         | $\times \mathbf{1.226}$ |
| SeisSol static excess<br>( $f_0 - \mu_{\text{app}} \cdot \sigma_n = 79.0$ MPa) | $\times 0.949$ (below) | $\times \mathbf{1.076}$ |

**Caveat — MFEM's gate is stricter than SeisSol's, and 85 MPa is marginal by it.** The `np=4` dry-run on the revised config (2026-07-09) reports:

```
[derived] nucleation overshoot (|tau_pre + delta_tau| - f_ss*sigma_n) = -3.56
↳ MPa (below steady-state strength)
[derived] PASS: rate-and-state initial conditions are well-posed.
```

MFEM compares  $|\tau_{\text{pre}} + \Delta\tau|$  against the *steady-state* strength  $f_{\text{ss}} \cdot \sigma_n$ , and at  $V_{\text{init}} = 1e-12$  the rate/direct effect lifts  $f_{\text{ss}} \approx 0.655 > f_0 = 0.6$ , so MFEM's threshold is  $\sim 8$  MPa above SeisSol's static excess. That is the whole discrepancy: 85 MPa clears SeisSol's criterion ( $\times 1.076$ , and SeisSol *did* nucleate) but sits  $-3.56$  MPa under MFEM's. **User decision (2026-07-09): keep 85 MPa** — match the SeisSol-run forcing and treat the overshoot line as the conservative static heuristic it is (the driver still reports well-posed; only the time-stepping settles nucleation). MFEM's own gate breaks even near 89 MPa (slope 0.987 MPa/MPa) if a later run shows only creep. Do **not** touch `radius_m` or the hypocentre.

Also quantify G10 here: report  $\max |r_{\text{MFEM}} - r_{\text{3D}}|$  over bump-support DOFs; if it exceeds  $\sim 1\%$  of  $R$ , the effective bump is measurably wider than SeisSol's.

5. **np=4 smoke, 200 steps, local.** Watch the R-020 `rate_state` at `np>1` warning (G11) — it

fires on every SAFS RS run today; confirm it is the ADER-substep path and not the RK path, or accept it.

6. **Short Expanse pilot (user approval required).** `--tfinal 5s, 2N×100r, --paraview-fault-dt 0.5s`. Confirm: rupture nucleates in the bump, no NaN, no free-surface daylighting blow-up,  $V_{\max}$  peaks and decays. Wall-time per simulated second  $\times 150$  gives the real budget.
  - 6b. **p2 convergence window (D2).** Same config with `[mesh].order = 2, tfinal = "20s"`. Compare the rupture-front position and peak slip rate against the p1 run at  $t \in \{4, 8, 12, 16, 20\}$  s. Agreement within the snapshot cadence  $\Rightarrow$  p1 resolves the cohesive zone; disagreement  $\Rightarrow$  report the p1 150 s result as qualitative only.  $\sim 4\times$  the p1 cost per simulated second.
7. **Production, checkpoint-chained.** Run the **parity arm** (`sigma_n_strength_floor_pa = -1.0`) first. Inspect the first two fault snapshots for the trace-cell runaway signature before committing the full wall-time; switch to the `snfloor10` arm only if it appears. Compare against `~/Downloads/SAFS_V3.4.1_..._outputs/`: fault-surface `SRs/SRd/Sls/Sld/Ts0/Td0/Pn0` at matched  $t \in \{4, 8, \dots, 96\}$  s. Reference data stops at  $t \approx 98$  s. Key metrics: rupture-front arrival along the corridor arclength  $s$ , peak slip rate, whether the front crosses the San Gorgonio gate (PREFERRED corridor `kbar_min = 0.98–0.99`, barely above the 0.9 floor — the run decides), final slip,  $M_w$ .

Use `safs/seisol_quakeworx/toolbox/combined_workflow/postprocess.ipynb` for the SeisSol side — it already reads any SeisSol output directory and emits  $M_w$  / slip / stress-drop / finished?.

### 1.13 Risks

| Risk                                                                                                                                                                                                                         | Mitigation                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nucleation margins are <math>\times 1.04</math> (binding).</b> MFEM's p1 discretization resolves the 583 m supra-S_E core with $\sim 1.5$ median fault edges; the bump may fail to nucleate where SeisSol's p3 succeeded. | Gate 4. Escalate <code>delta_tau_pa</code> $\rightarrow$ 85e6 per the deck's own guidance.                                                                                                |
| <b>Trace-cell runaway with <code>sigma_n_strength_floor_pa = -1</code> (D1).</b>                                                                                                                                             | Run the 10e6 arm in parallel; <code>project_pre_k25_case2_runaway_slip_not_mesh</code> documents the signature (isolated cells sliding $\sim 4$ m/s forever at $\sigma_n$ 0.1–5 MPa).     |
| <b>GTS cost.</b> MFEM has no LTS; SeisSol's $30.5\times$ LTS speedup is unavailable.                                                                                                                                         | Phase 1 measures it. Checkpoint-chain the production run. The mesh is <i>smaller</i> than what MFEM already runs at $2N\times 100r$ , so this is a budget problem, not a feasibility one. |
| <b>Mixed flux is not SeisSol's flux.</b> Central-on-fault-adjacent faces vs pure Godunov. Differences in the rupture front are the <i>point</i> of the experiment, not a bug.                                                | Also run the <code>--mixed-flux none</code> <code>--time-integrator ader</code> arm (the upwind sibling <code>sbatch</code> ) so the flux is the only variable.                           |

| Risk                                                                                                                                                                           | Mitigation                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>p1 vs p3.</b> Under-resolved cohesive zone $\Rightarrow$ artificially high peak slip rate / delayed front.                                                                  | D2; p2 convergence check on a 20 s window.                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Doc rot.</b> CLAUDE.md and spatial_friction_config_schema.md both make stale claims about spatial-driver nucleation. An implementer following them will do the wrong thing. | Phase 3 fixes both.                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Sidecar memory (G12).</b> The three sidecars are replicated per rank and total 1329 MB. At 2 GB/rank the job likely OOMs after $\sim 20$ min of setup.                      | The new sbatch defaults to <code>--mem=200000M</code> (4 GB/rank; compute nodes are exclusive and charged whole-node, so this is free) and carries a pre-flight that recomputes the ratio from <code>SLURM_MEM_PER_NODE / SLURM_NTASKS_PER_NODE</code> and errors above 90%. If memory ever gets tight, crop the CVM sidecar to the mesh bbox — that alone removes $\sim$ half of <code>velocity_safs.h5</code> . |
| <b>Stale-ABI builds (G13).</b> Any future change to a struct in <code>spatial_friction.hpp</code> silently corrupts incrementally-built binaries.                              | Fixed: <code>SPATIAL_HEADERS</code> added to <code>SEAS_HEADERS</code> , plus an explicit prerequisite on the <code>fault_face_flux.o</code> rule (which uses <code>DYNAMIC_HEADERS</code> ). Verified by touching the header and confirming all seven dependent TUs recompile.                                                                                                                                   |

## 1.14 Summary of files

### Create (6):

`safs/project_7.0_preferred/friction/code/build_pref_friction_sidecar.py`  
 uses the shared `velocity/code/sidecar.py::write_sidecar`; no hand-rolled HDF5

`safs/project_7.0_preferred/config/`  
`spatial_friction_safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2.toml`  
 parity arm, `sigma_n_strength_floor_pa = -1.0`  
`spatial_friction_safs_seisol_v3_4_1_RSSRW_PREFERRED_THERMAL_CASE2_snfloor10.toml`  
 fallback arm; differs only in that key + `output_dir`

`seas_test_spatial_friction_sidecar/`                      3 tests  
 + the `f_w = 0` parser/psi tests from Phase 2

`jobs/safs/safs_expanse/`  
`spatial_dyn_ratestate_v3_4_1_pref_thermal_case2..._2hr_safs.sbatch` (+ restart  
 $\hookrightarrow$  sibling)

### Change (6):

|                                                                         |                                                               |
|-------------------------------------------------------------------------|---------------------------------------------------------------|
| spatial/code/spatial_friction.hpp                                       | FrictionSidecarSpec, RateStateSidecarFields,<br>resolver ctor |
| spatial/code/spatial_friction.cpp                                       | :713 f_w >= 0 guard; parse + resolve the sidecar              |
| drivers/spatial_dyn_driver.cpp                                          | load sidecar fields, extend --print-derived                   |
| safs/project_7.0_preferred/velocity/code/build_pref_velocity_sidecar.py |                                                               |
|                                                                         | argparse; stale hard-coded path                               |
| safs/project_7.0_alternative/document/spatial_friction_config_schema.md |                                                               |
|                                                                         | new block; fix stale nucleation claims                        |
| miniapps/seas/CLAUDE.md                                                 | fix stale "single nucleation kind" paragraph                  |

**Re-run, no change (2):** - csm\_stress\_nc\_to\_mfem\_hdf5.py → the Andersonian k=1.8 sidecar -  
msh\_to\_puml.py — not needed; the mesh already exists on both sides

**Zero work:** mesh, boundary attrs, flux/integrator wiring, FL=103 friction law, compact-bell nucle-  
ation, stress reader, material reader, checkpoint/restart, ParaView fault + free-surface output.